**HOW TO FILL OUT THIS SHEET:**

- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

LAST NAME

The image shows two 6x6 letter grids for a word search. The left grid is titled 'DEBBIE' and the right grid is titled 'BACH'. Both grids contain letters and black squares representing obstacles.

DEBBIE Grid:

D	E	B	B	I	E
A	A	A	A	A	A
H	B			B	H
C	C	C	C	C	
B	B	B			
E		E			

BACH Grid:

B	A	C	H		
A		A	A		
	B	B	B	H	
C	C				
D	D				
E					

EXAMPLE 2: Chan Ai Beng

FIRST NAME

LAST NAME

C	H	A	N
A	A		A
B	B	B	B
	C	C	
D			
E			

A	I	B	E	N
	A	A	A	A
B	B	B		B
C	C	C	C	C
D	D	D	D	D
E	E	E	E	E

EXAMPLE 3: Jamal bin Abas

FIRST NAME

LAST NAME

J	A	M	A	L		B	I	N	A	B	A	S
A	●	A	●	A	●	A	●	A	●	●	A	●
B	●	B	●	B	●	B	●	B	●	●	B	●
C	●	C	●	C	●	C	●	C	●	●	C	●
D	●	D	●	D	●	D	●	D	●	●	D	●
E	●	E	●	E	●	E	●	E	●	●	E	●

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

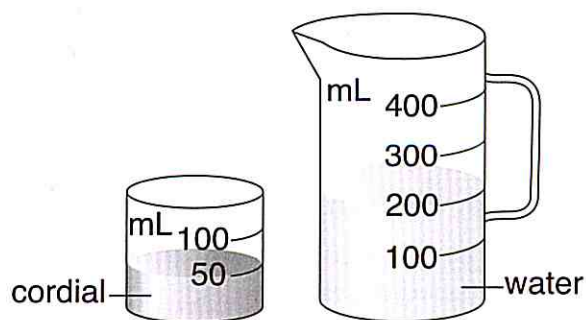
CLASS
(optional)

(A)	(K)
(B)	(L)
(C)	(M)
(D)	(N)
(E)	(O)
(F)	(P)
(G)	(Q)
(H)	(R)
(I)	(S)
(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

(A) (B) (C) **(D)**



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)
- 11 (A) (B) (C) (D)
- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)



S

2013 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	D	D	D	B	C	B	D	C	C
2	C	D	C	C	A	D	D	A	D
3	D	B	B	C	D	C	B	B	A
4	B	A	B	B	B	B	B	B	B
5	D	D	D	C	D	C	C	A	B
6	C	D	C	B	D	B	A	C	A
7	A	A	A	D	B	A	C	D	D
8	D	D	C	A	C	D	B	B	D
9	C	A	D	D	B	D	D	C	A
10	C	C	D	B	C	C	A	A	B
11	A	D	A	A	B	B	C	D	A
12	B	A	B	B	D	D	C	A	A
13	C	D	A	C	A	D	A	D	A
14	B	B	D	A	C	C	B	B	B
15	D	C	D	B	D	A	D	B	D
16	C	C	D	C	A	C	D	A	C
17	B	D	A	B	C	D	D	C	C
18	A	D	C	B	B	C	D	C	D
19	A	B	A	C	D	B	B	C	A
20	D	B	C	D	D	D	D	C	C
21	C	C	C	D	C	B	C	A	D
22	C	C	A	D	B	D	A	D	D
23	C	C	A	D	C	D	C	D	D
24	B	D	A	D	B	B	C	C	D
25	B	B	B	C	D	B	D	A	C
26	A	A	A	C	C	C	D	C	D
27	A	A	D	A	A	C	B	B	C
28	B	D	C	A	D	D	C	D	A
29	D	C	A	C	C	A	C	A	D
30	D	A	C	D	C	C	A	C	A

(Please turn over)



S

2013 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			C	B	B	D	C	B	C
32			A	C	C	C	B	C	B
33			C	D	D	C	D	C	C
34			B	D	D	C	B	D	D
35			D	D	D	C	B	C	C
36			C	C	A	A	B	A	D
37			D	B	C	D	B	D	B
38			B	D	C	C	A	B	C
39			B	A	A	C	A	C	D
40			B	A	D	D	C	B	D
41					A	D	D	B	A
42					A	D	C	B	A
43					D	A	B	A	D
44					C	C	C	D	B
45					B	B	D	D	A

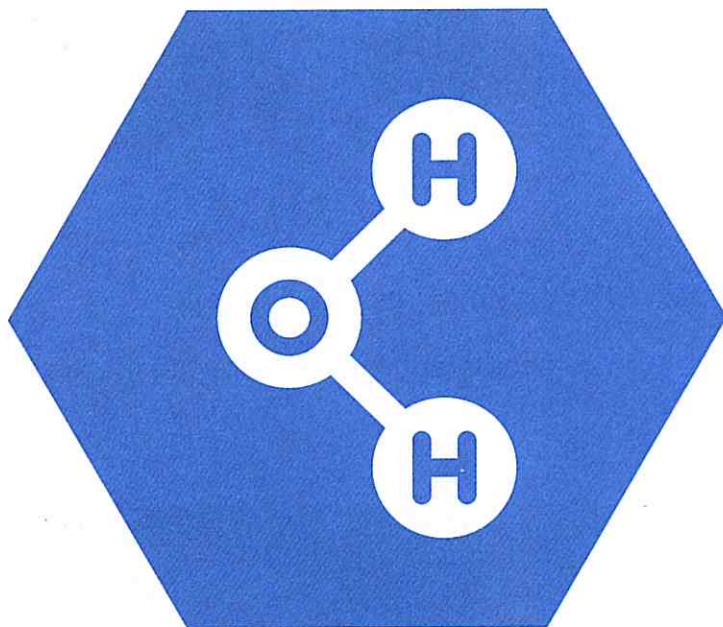
Contact Details

Telephone: (02) 8344 1016
Email: info@eaa.unsw.edu.au
Website: www.eaa.unsw.edu.au



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER B



2013 ICAS

International Competitions
and Assessments for Schools

SCIENCE

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Stan collected some rocks on his last holiday.

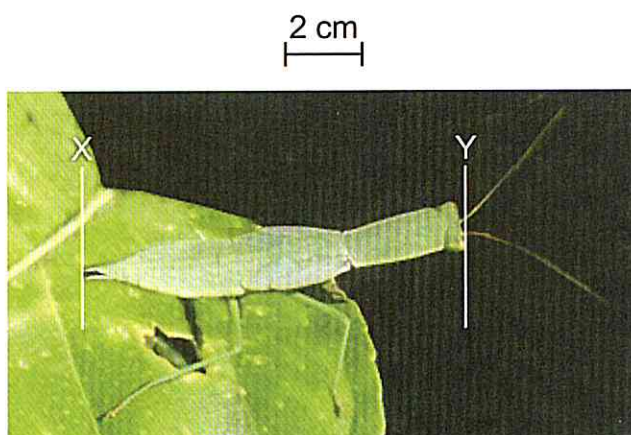
His favourite rock is the one that is smooth, rounded and has a light stripe.

Which of these rocks is Stan's favourite?



2. The praying mantis is an insect.

The photograph shows a praying mantis on a leaf.



What is the length (in cm) of the praying mantis from X to Y?

- (A) 2
(B) 4
(C) 5
(D) 10

3. For a rocket to be launched and leave a planet, it must reach a certain speed. This speed is called the escape velocity.

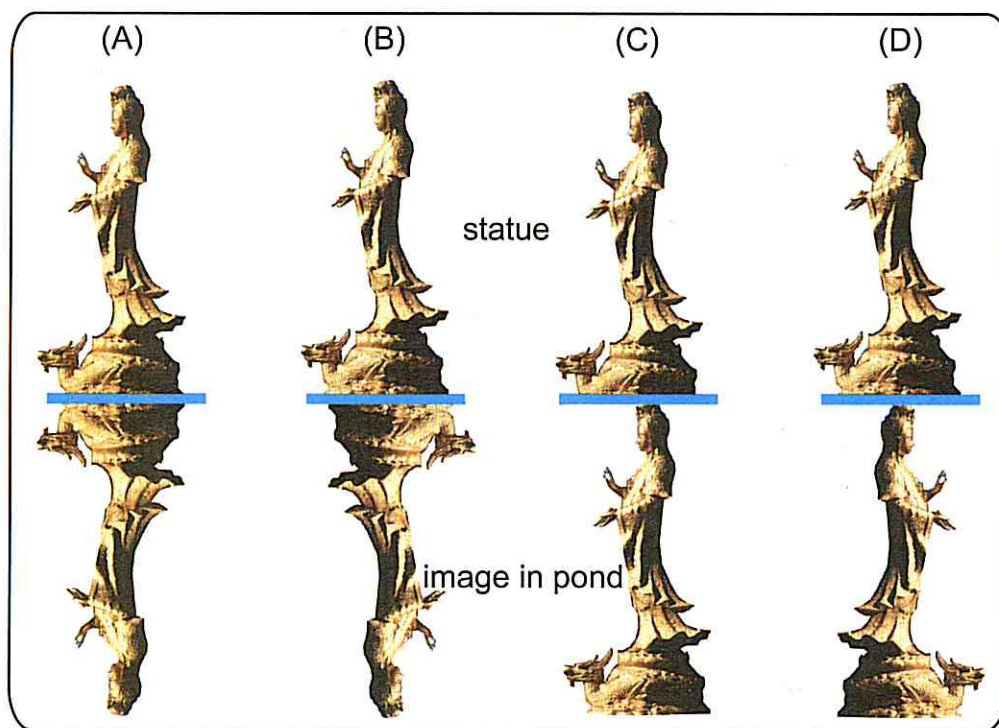
The table shows the escape velocity of each planet in the solar system.

Planet	Escape velocity (km/s)
Mercury	$4 \frac{1}{2}$
Venus	$10 \frac{1}{2}$
Earth	11
Mars	5
Jupiter	$59 \frac{1}{2}$
Saturn	$35 \frac{1}{2}$
Uranus	$21 \frac{1}{2}$
Neptune	$23 \frac{1}{2}$

How many planets have an escape velocity greater than that of Earth?

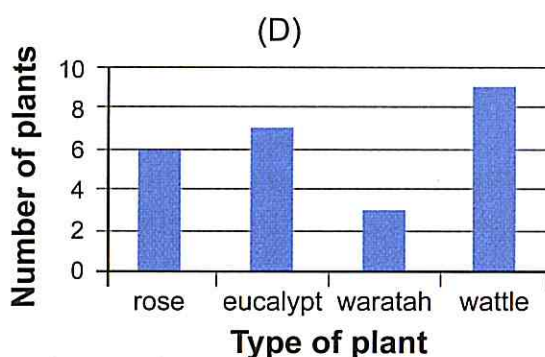
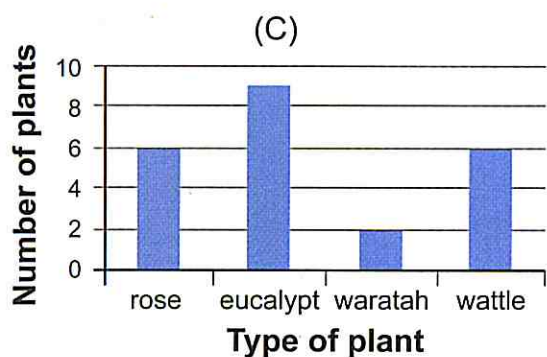
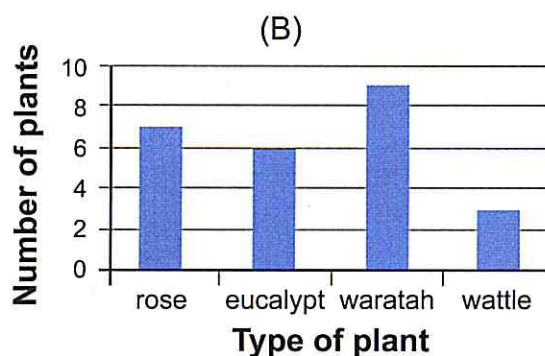
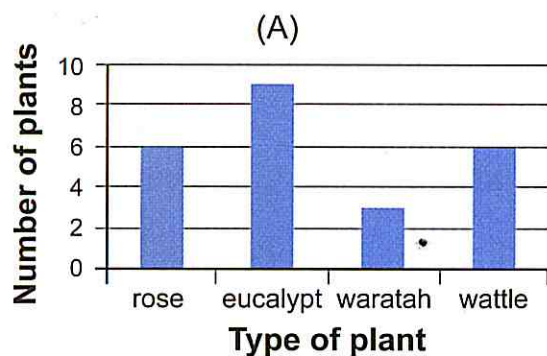
- (A) 3
(B) 4
(C) 5
(D) 6

4. Nick observed a statue standing in the middle of a pond and its reflection on the water. Which diagram shows the statue and its reflected image correctly?



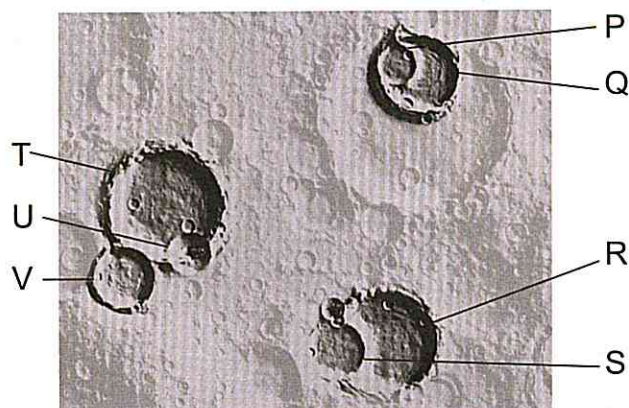
5. As Joe walked through a garden, he saw six rose bushes, seven eucalypt trees, three waratah shrubs and nine wattle trees.

Which graph matches Joe's observations?



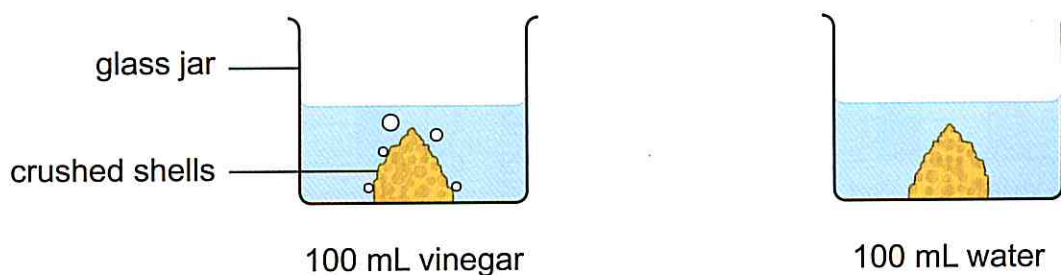
For questions 6 and 7 use the information below.

The diagram represents some craters on the surface of the Moon.



6. Which of these craters is the widest?
- (A) Q (B) R (C) S (D) T
7. Which statement is supported by the information in the diagram?
- (A) Crater R was formed before crater S.
(B) Crater Q was formed before crater R.
(C) Crater V was formed before crater T.
(D) Crater U was formed before crater V.

8. Simon predicted that seashells would dissolve in vinegar but not in water.
He set up an investigation using crushed shells.
Simon drew these two diagrams to show his teacher what happened in his investigation.

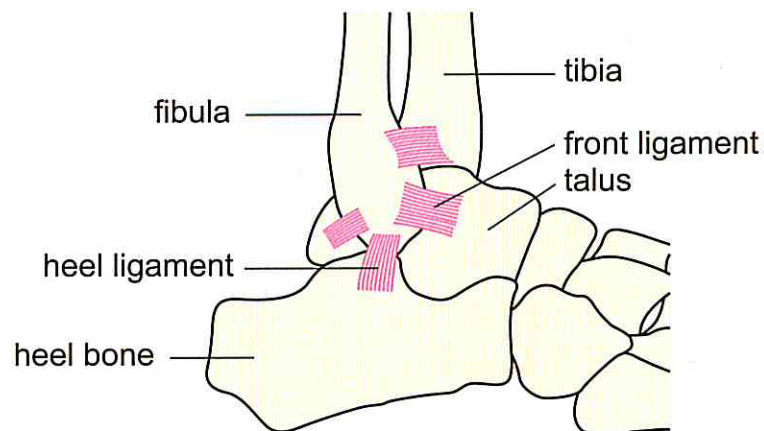


Which of the following is an observation that Simon recorded in his diagrams?

- (A) The vinegar became warm.
(B) The crushed shells dissolved in the vinegar.
(C) The crushed shells in both jars made bubbles.
(D) The crushed shells in the vinegar made bubbles.

9. Ligaments hold bones together.

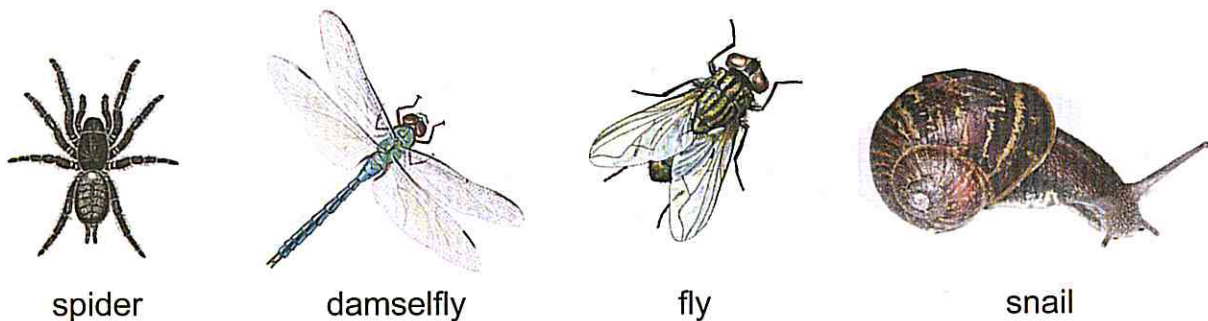
The diagram shows some ligaments and bones of the ankle.



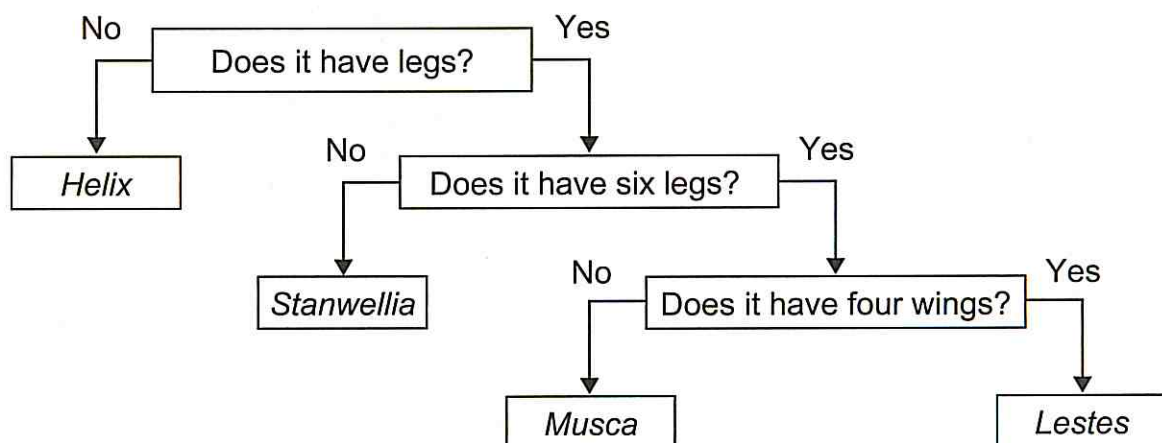
Which two bones are connected by the front ligament?

- (A) fibula and talus
(B) tibia and fibula
(C) fibula and heel bone
(D) tibia and heel bone

10. Some students took these photographs.



They used the key below to find the scientific name of the fly.

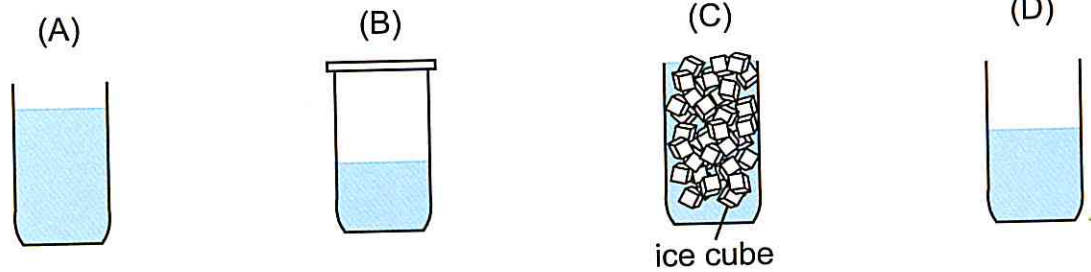


What is the scientific name of the fly?

- (A) *Helix* (B) *Stanwellia* (C) *Musca* (D) *Lestes*

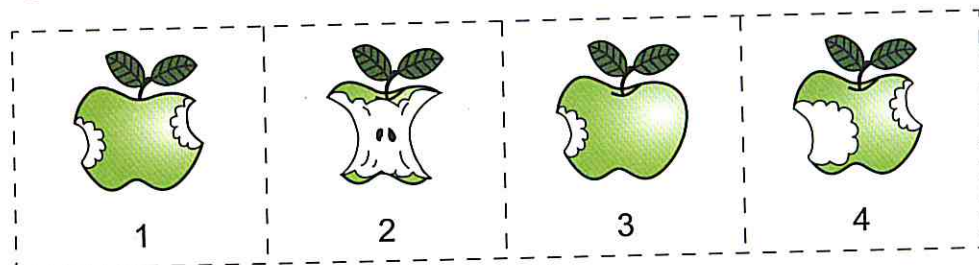
11. Sue set up four identical containers as shown below. She placed the containers close together in a warm spot on a window sill.

Which container will be the first to lose all of its water due to evaporation?



12. Belle ate an apple. She drew the apple at four stages as she was eating it.

Her drawings are shown but not in the correct order.

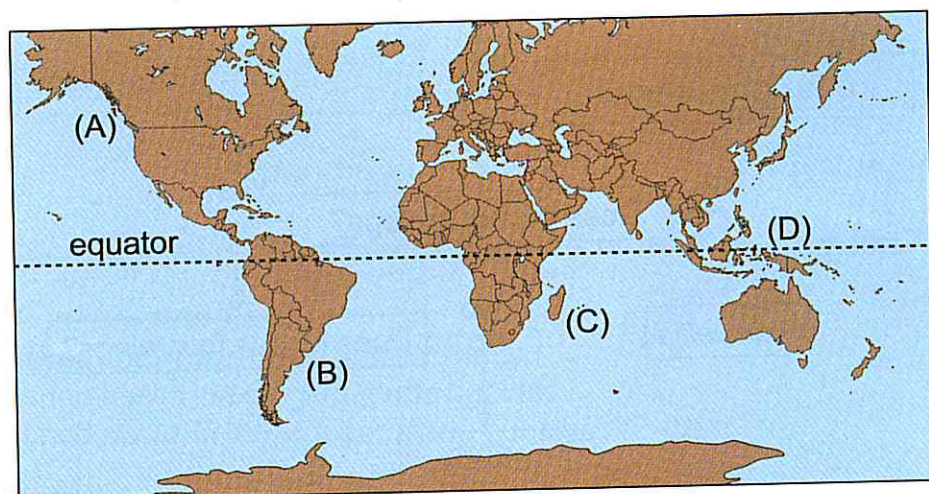


What is the correct order of the drawings to show the apple as she ate it?

- (A) 3, 1, 4, 2 (B) 1, 3, 4, 2 (C) 3, 4, 1, 2 (D) 2, 4, 1, 3

13. Tropical cyclones form only in regions where the ocean water is warm and the air is warm.

Which point on this map shows where tropical cyclones are most likely to form?



14. The table provides information about some of the planets in our solar system.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus
Diameter (km)	4 879	12 104	12 756	6 792	142 984	120 536	51 118
Distance from Sun (millions km)	58	108	150	228	779	1433	2872
Time to go around the Sun once	88 days	225 days	365 days	687 days	12 years	29 years	84 years
Time to spin on axis once	59 days	243 days	24 hours	25 hours	10 hours	11 hours	17 hours

Which two planets are most similar in size?

- (A) Mercury and Mars
(B) Venus and Earth
(C) Earth and Mars
(D) Jupiter and Saturn

15. Cali wanted to find out what substances look like when they are mixed with water.

The table lists her results.

Substance mixed with water	What colour is the mixture?	Can you see through the mixture?
milk	white	no
mud	brown	no
salt	no colour	yes
sugar	no colour	yes
tea	brown	yes

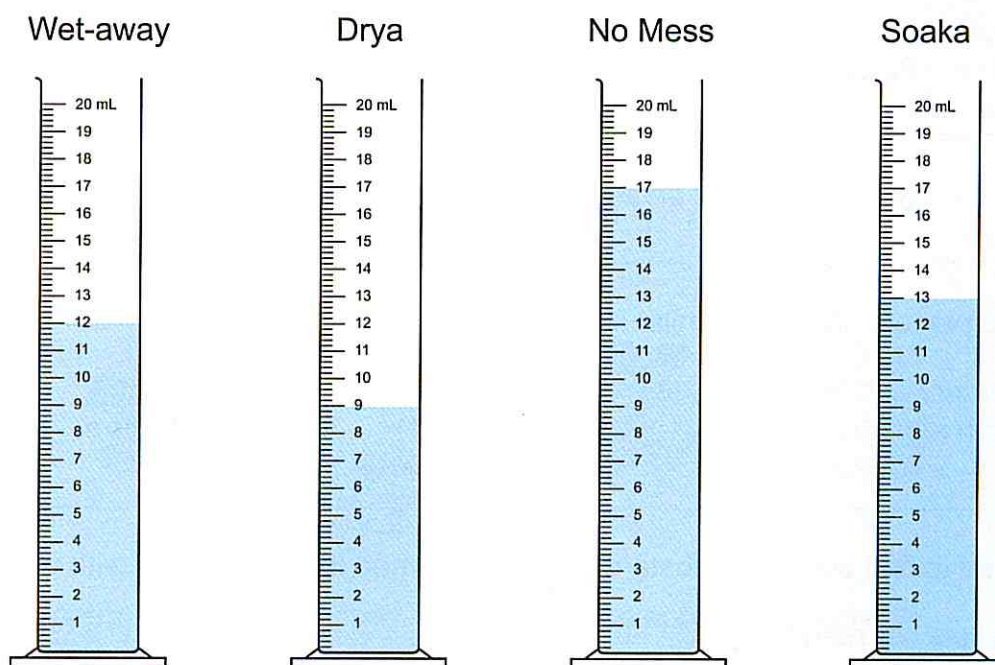
Which of these did Cali have to keep the same to be able to compare her results?

- (A) the spoon she used to stir the mixture
(B) the size of the glass to hold the mixture
(C) the amount of substance added to the water
(D) the colour of the substance added to the water

For questions 16 and 17 use the information below.

Sharon and Andrew wanted to compare four different brands of paper towel. They had equal-sized paper towels of four different brands: Wet-away, Drya, No Mess, Soaka. They had four measuring cylinders, one for each paper towel. Each measuring cylinder contained 20 mL of water.

Sharon and Andrew slowly poured water onto each paper towel until the paper towel could not soak up any more. The diagrams show the water left in the measuring cylinders after they had stopped pouring water onto the paper towels.



16. Which paper towel soaked up the **LEAST** amount of water and how much did it soak up?

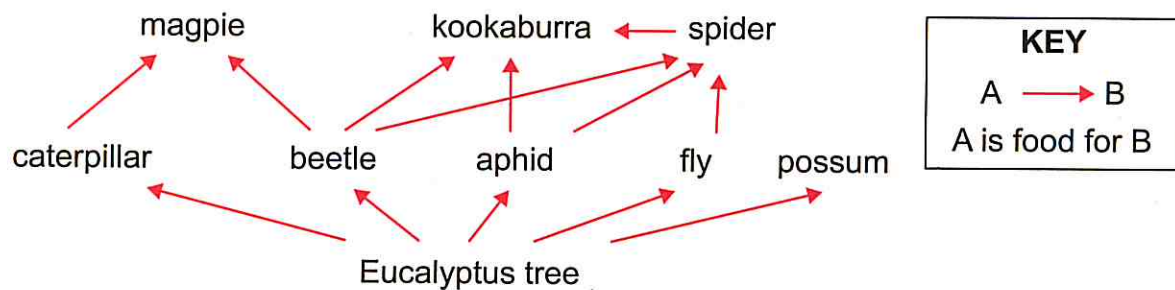
	Paper Towel	Amount of water soaked up (mL)
(A)	Drya	9
(B)	Drya	11
(C)	No Mess	3
(D)	No Mess	17

17. Was this a fair test?

- (A) No, because each paper towel soaked up different amounts of water.
- (B) Yes, because they used the same amount of water each time.
- (C) No, because the paper towels were all different brands.
- (D) Yes, because each paper towel was the same size.

18. Plants are called producers because they make food using energy from the Sun. An organism that eats another organism is called a consumer. A consumer that eats a producer is called a first-order consumer.

The diagram shows a food web.



How many first-order consumers are shown in this food web?

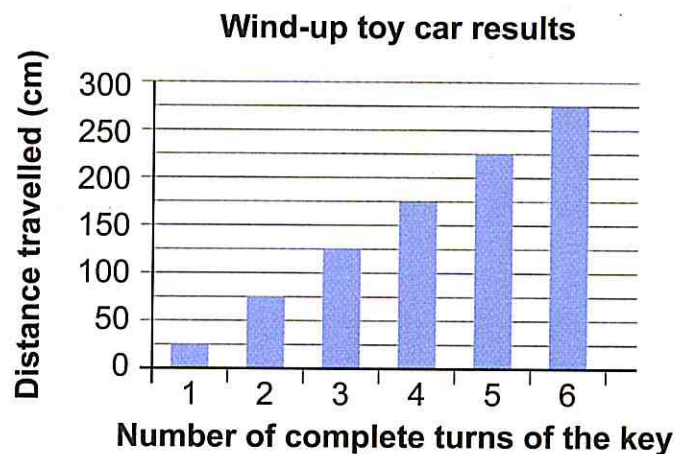
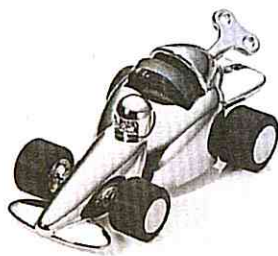
- (A) 1 (B) 3 (C) 4 (D) 5

For questions 19 and 20 use the information below.

Wind-up toy cars move forwards when you wind them up with a key and let them go.

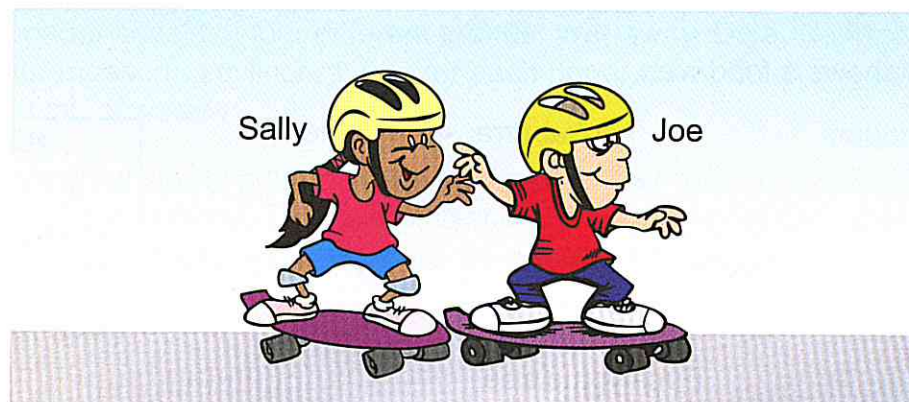
Frank carried out an investigation to find out if the number of times he turns the key affects the distance the car travels.

The graph shows Frank's results.



19. What is the least number of complete turns of the key needed to make the car travel 375 cm?
- (A) 7 (B) 8 (C) 9 (D) 10
20. Which two things did Frank have to keep the same to make his investigation fair?
- (A) the car and the distance it travelled
 (B) the car and the surface on which it moved
 (C) the distance the car travelled and the surface on which it moved
 (D) the number of turns of the key and the surface on which the car moved

21. Sally and Joe are standing on their skateboards, on a smooth flat surface. They are facing in the same direction. Sally and Joe have the same body mass.



Sally gently pushes Joe and they both start moving.

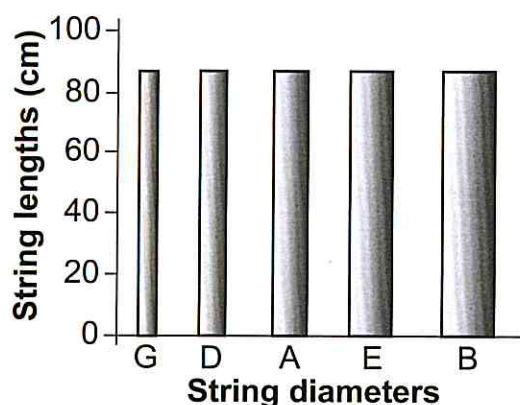
In which directions do Joe and Sally move?

	Sally moves	Joe moves
(A)	forward	forward
(B)	forward	backward
(C)	backward	forward
(D)	backward	backward

22. A guitar produces sound when its strings vibrate.

The number of times a string vibrates in one second is called its frequency. Frequency is measured in hertz (Hz). The greater the frequency, the higher the pitch of the sound produced.

The graph and table give information about the sound produced by a 5-string bass guitar.



String	Frequency (Hz)
G	98
D	73.5
A	55
E	41
B	30.9

What can be inferred from this information?

- (A) Shorter strings produce lower pitched sound.
- (B) Shorter strings produce higher pitched sound.
- (C) Thicker strings produce lower pitched sound.
- (D) Thicker strings produce higher pitched sound.

For questions 23 and 24 use the information below.

Different animals have different hearing ranges. An animal's lower limit of hearing is the lowest frequency sound it can hear. Its upper limit of hearing is the highest frequency sound it can hear. The hearing range of an animal is made up of all the sound frequencies in between.

The table shows information about some animals.

Animal	Lower limit of hearing (Hz)	Upper limit of hearing (Hz)
bird	250	21 000
crocodile	20	6 000
dog	15	50 000
bat	1 000	150 000
African moth	5 000	110 000
dolphin	150	150 000

23. How many of these animals can hear a sound with a frequency of 125 Hz?
(A) 0 (B) 1 (C) 2 (D) 3
24. Which animal has the largest hearing range?
(A) dog (B) bat
(C) African moth (D) dolphin

25. The table shows how many nuts of the same type make up 100 grams.

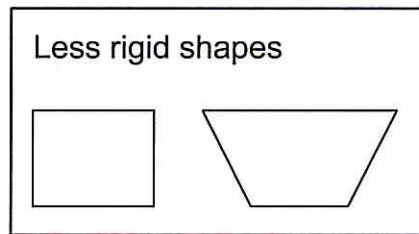
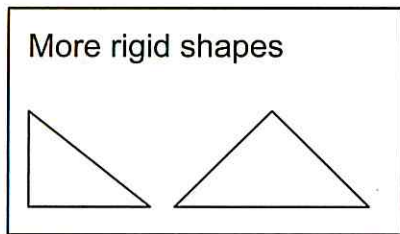
Type of nut	Number of nuts in 100 grams
almond	81
brazil nut	21
hazelnut	74
macadamia nut	39
pecan nut	67
pine nut	588
pistachio nut	173

Brad weighed one nut of each type.

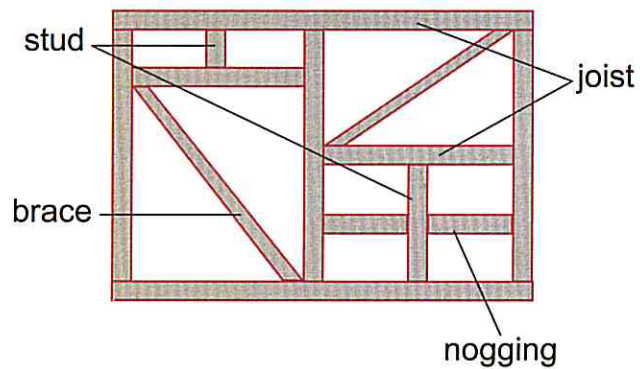
Which nut would have the greatest mass?

- (A) almond
(B) brazil nut
(C) pine nut
(D) pistachio nut

26. Something that is rigid is hard to push out of shape. Some shapes are more rigid than others.



Bob built a wooden frame from straight pieces of wood as shown.

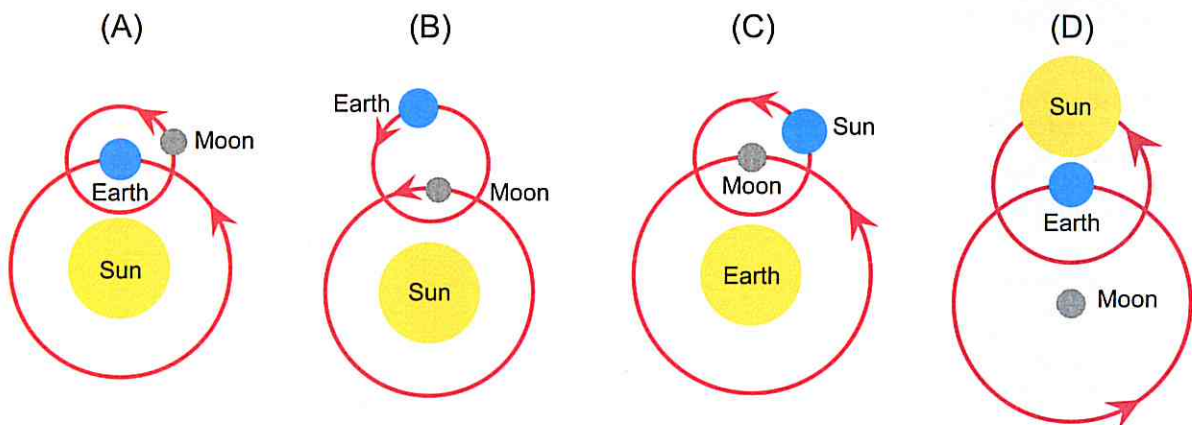


Which piece of wood was added to the frame to make it more rigid?

- (A) brace (B) joist (C) nogging (D) stud

27. Earth moves around the Sun. As Earth moves around the Sun, the Moon moves around Earth.

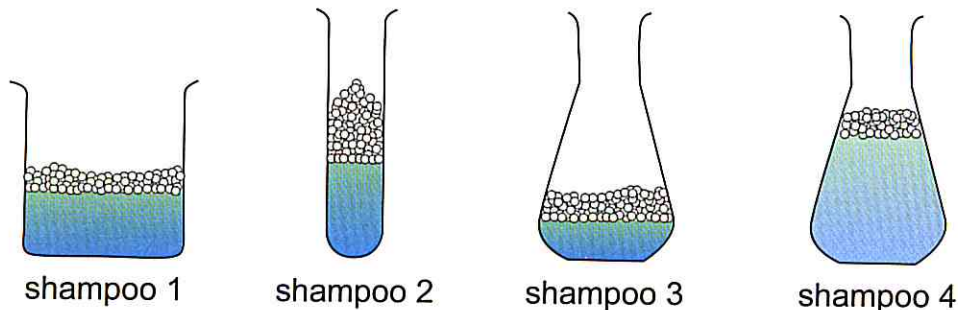
Which diagram best represents the movement of Earth and the Moon?



28. Some students wanted to test which brand of shampoo produced the most bubbles.

They added some water to four glass containers. Next they added three drops of shampoo to each container. Each container had a different shampoo. The students covered each container and shook each one for the same length of time.

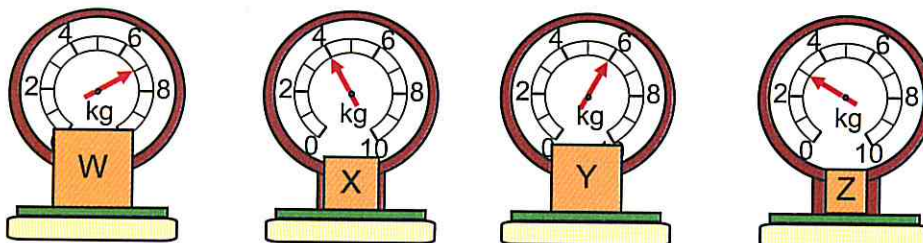
The diagrams show what their containers looked like after shaking.



Why was this investigation **NOT** a fair test?

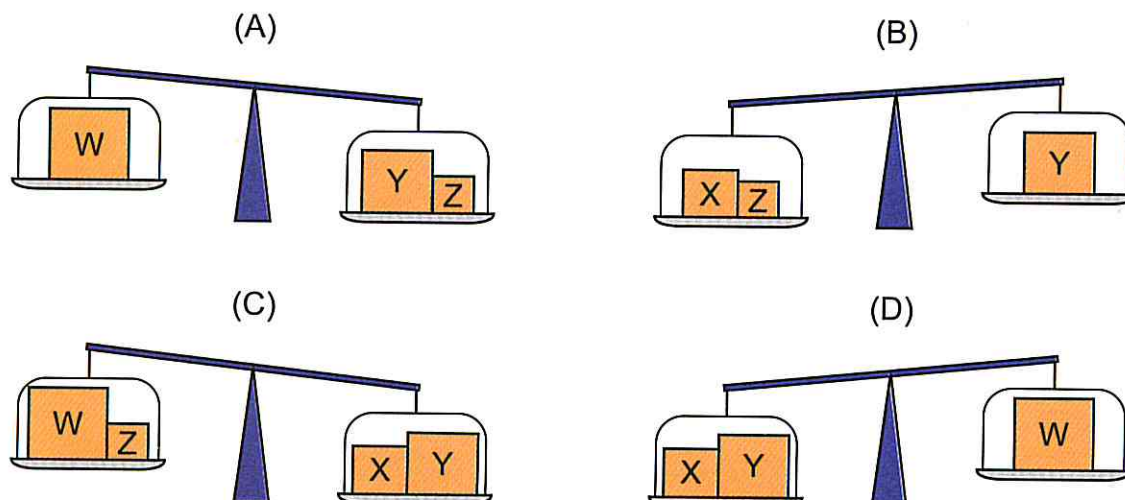
- (A) Each container was made of glass.
- (B) Each container had a different amount of bubbles in it.
- (C) Each container had only three drops of shampoo added.
- (D) Each container had a different amount of water added to it.

29. Frans used scales to find the mass of each of four different objects.



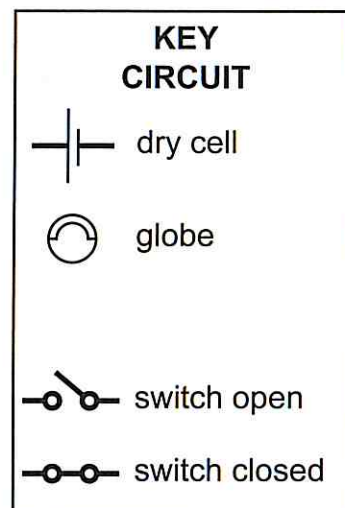
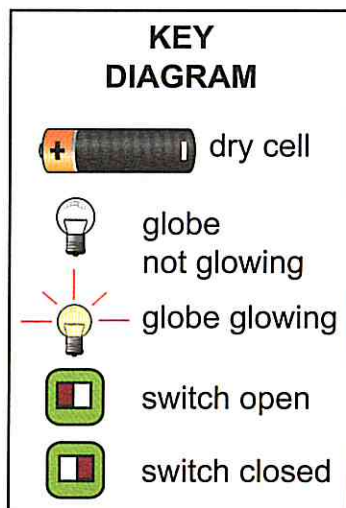
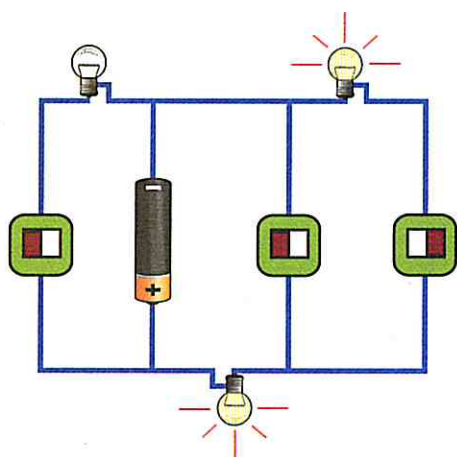
He then compared the masses of the objects using a balance.

Which diagram shows the balance comparison that is **NOT** correct?

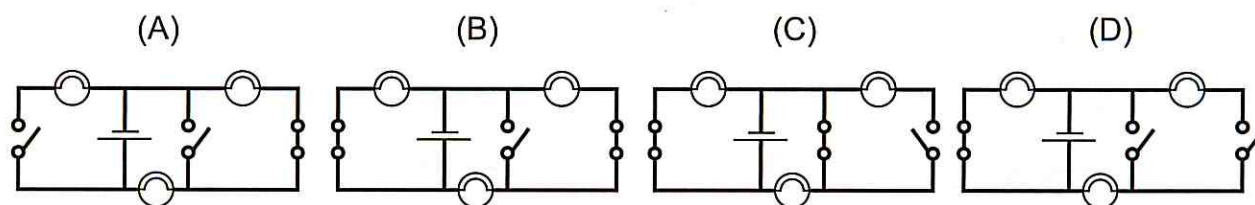


30. Electricity can flow in a circuit when there is an unbroken path for the electricity to follow. An open switch breaks the path.

The diagram shows an electric circuit. All of the globes have been tested and they work.



Which circuit matches the diagram shown above?



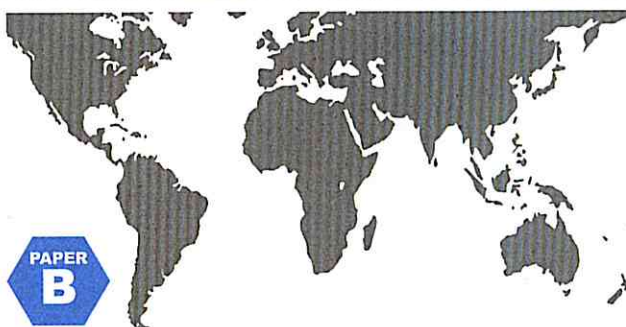
Acknowledgment

Copyright in this booklet is owned by Educational Assessment Australia, UNSW Global Pty Limited, unless otherwise indicated. Every effort has been made to trace and acknowledge copyright. Educational Assessment Australia apologises for any accidental infringement and welcomes information to redress the situation.



The following year levels should sit **THIS Paper**:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 5
Singapore	Primary 3
South Africa	Grade 4



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**Educational
Assessment
Australia**
eaa.unsw.edu.au

© 2013 Educational Assessment Australia.
EAA is an education group of UNSW Global Pty Limited, a not-for-profit provider of education, training and advisory services and a wholly owned enterprise of the University of New South Wales. ABN 62 086 418 582